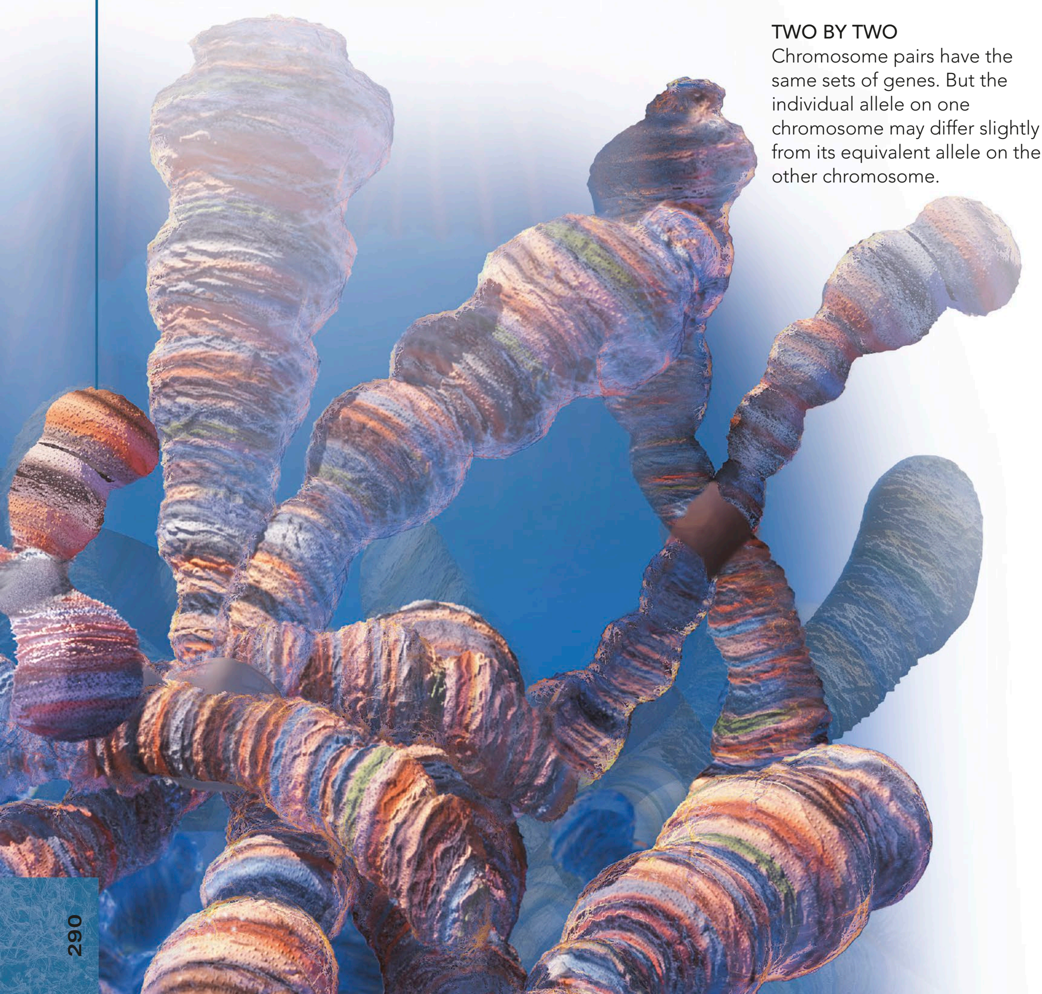
PATTERNS OF INHERITANCE

GENES ARE PASSED FROM ONE GENERATION TO THE NEXT, IN A VAST SEQUENCE OF INHERITANCE. THEY ARE RESHUFFLED AT EACH STAGE SO THAT OFFSPRING ARE UNIQUE, BUT THERE ARE PATTERNS IN THE MODE OF INHERITANCE.

VERSIONS OF GENES

Each cell in a body contains a double-set of genetic material, in the form of 23 pairs of chromosomes. One chromosome of each pair, and the genes on it, come from the mother. The other chromosome is from the father. So there are, in effect, two versions of every gene in the set – one maternal and one paternal. These versions of genes are called alleles. Inheritance patterns vary depending on how these two versions interact, because they may be identical or slightly different.



TWO BY TWO
Chromosome pairs have the same sets of genes. But the individual allele on one chromosome may differ slightly from its equivalent allele on the other chromosome.